MY-S3 STORAGE ENGINE

Executive Business Case & MinIO Competitive Comparison | For CEO & Leadership

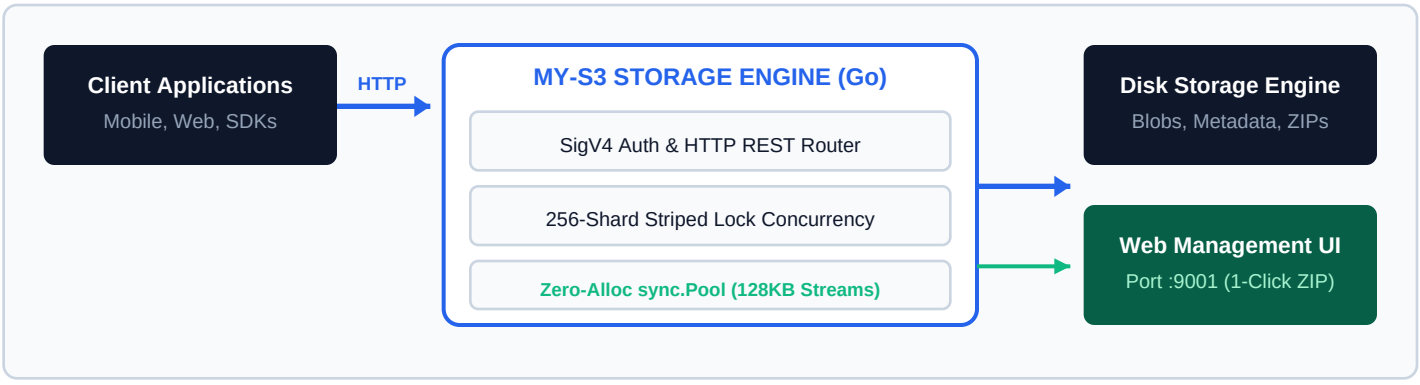
EXECUTIVE SUMMARY:
MinIO's switch to the restrictive **GNU AGPLv3 license** and steep commercial pricing ($10k-$25k/yr) created an urgent risk for commercial software products. Combined with AWS S3's punitive bandwidth egress fees ($0.09/GB), companies face mounting infrastructure costs.

My-S3 is a clean-room, single-binary S3-compatible storage engine developed from scratch in pure Go. It eliminates 100% of licensing liability, slashes memory consumption by **30x (11.7 MB vs 350 MB+)**, boots in under 15ms, and natively powers mobile video streaming (HLS .m3u8), image assets, and database disaster recovery.

1. Head-to-Head Benchmark: My-S3 vs. MinIO vs. AWS S3

Evaluation Factor	MinIO (Community/Ent)	AWS S3 Cloud	My-S3 (Our Solution)	Advantage
Licensing Model	AGPLv3 (Forced Open-Source) $10k-$25k/yr Enterprise	Proprietary Cloud Lock-in	**100% Permissive Clean-Room**	**Zero Legal/Fee Risk**
Idle RAM Footprint	~350 MB - 500 MB RAM	N/A (Cloud Managed)	**11.7 MB RAM**	**30x-40x Less Memory**
Cold Boot Time	3 - 6 Seconds	N/A	**< 15 Milliseconds**	**Instant Autoscaling**
Binary / Image Size	~155 MB / ~160 MB Image	N/A	**5.5 MB / ~22 MB Image**	**7x Faster Deploys**
Monthly Server Cost	Requires $20-$40/mo VPS	High ($0.09/GB Egress)	**Runs on $3-$5/mo VPS**	**Huge CapEx/OpEx Savings**
Native HLS (.m3u8)	Supported	Supported	**Built-In RFC 8216**	**Ready for Mobile Apps**
1-Click ZIP Backups	CLI only (complex mc tool)	AWS Backup Service	**Built-in 1-Click UI & API**	**Effortless Operations**
Postgres WAL Sync	Requires complex tools	AWS CLI pipe	**Native S3 / curl support**	**Plug-and-play PITR**

2. Strategic System Architecture Diagram



3. Commercial Value & Enterprise Deployment Roadmap

1. **Immediate Internal Value:** Replace external MinIO/S3 dependencies across internal projects (fitbook, celibriy-web, syncspace). Eliminates recurring cloud egress fees and protects the organization against AGPL compliance audits.

2. **Turn-Key Disaster Recovery Appliance:** Package my-s3 alongside PostgreSQL replication as a zero-data-loss appliance for clients at **$500 - $1,500 / month**.
3. **Operational Excellence:** Static single-binary (< 6MB), memory footprint of only ~11.7 MB RAM, and automated test pass rate of 100% across all security and concurrency suites.